

Test Plan: OrangeHRM Demo Automation

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AUT: opensource-demo.orangehrmlive.com

1. Objectives

1. Build a maintainable, reviewable UI automation suite covering the business-critical flows of five OrangeHRM modules: Login, Admin (user management), PIM (employee records), Leave, and Dashboard.
2. Implement the same test design in both Cypress and Playwright (JavaScript) to demonstrate tool-agnostic framework architecture (Page Object Model).
3. Provide fast feedback: a smoke suite that completes in ≤ 10 minutes on every PR, and a nightly full regression.
4. Produce professional deliverables (plans, designs, reports, traceability) equivalent to a real enterprise QA engagement.

2. Scope

2.1 In scope

MODULE	FLows COVERED
Login	Valid login, invalid credential matrix (wrong user / wrong password / both / empty fields), error-message assertions, forgot-password link navigation, logout, redirect protection for unauthenticated deep links
Admin	Search system users (by username, role, status), add user, edit user, delete user, record-count and grid assertions, validation messages on the add-user form
PIM	Add employee (basic + with login details), search employee (name / ID), edit personal details, delete employee, grid pagination/row assertions
Leave	Apply leave, leave list filtering (date range, status, type), verify applied leave appears in list
Dashboard	Post-login landing verification, presence of core widgets (Time at Work, My Actions, Quick Launch), quick-launch tile navigation

2.2 Out of scope

- Other OrangeHRM modules (Buzz, Recruitment, Performance, Time, My Info, Directory, Claim, Maintenance).
- API, performance, load, and security testing (UI functional automation only).
- Mobile/responsive layouts; accessibility audits (may be a later phase).
- Email flows (password-reset email cannot be received on the demo tenant).
- Localization; only English UI is tested.
- Testing OrangeHRM upgrades themselves — the demo version current at execution time is the baseline.

3. Test strategy

- **Approach:** Risk-based UI automation using the Page Object Model. Business-critical paths (login, employee CRUD) get both positive and negative coverage; peripheral flows get positive-path coverage only.

- **Tooling parity:** One set of manual test cases (single source of truth) implemented twice — Cypress and Playwright. Design, locators, and data rules are shared conceptually; implementations stay idiomatic to each tool.
- **Data strategy:** The demo environment is public and resets periodically, so tests never depend on pre-existing data except the Admin account. Every record a test needs is created by the test (or its setup) with a unique timestamp suffix, asserted on, and where practical deleted afterwards.
- **Stability rules:** No fixed sleeps; web-first/retryable assertions only; locator preference order defined in the HLD; each new test must pass 3 consecutive local runs before merge.
- **Defect management:** AUT defects and script defects are both logged as GitHub Issues, labelled aut-bug vs test-bug, with severity (Critical/High/Medium/Low) and evidence (screenshot/trace).

4. Test levels

LEVEL	APPLICABILITY HERE
Unit	Framework utilities only (e.g., data generators) — lightweight, optional
Integration	Page-object-level checks exercised implicitly through E2E flows
System (E2E)	Primary level — full user journeys through the browser
Smoke	Tagged subset (~1–2 cases per module) run on every PR
Regression	Full suite, nightly and pre-release

5. Environment details

ITEM	DETAIL
AUT URL	opensource-demo.orangehrmlive.com/web/index.php/auth/login
Account	Admin / admin123 (demo-published credentials; still injected via env vars, never hard-coded)
Environment type	Public shared demo, periodic data resets — treat as unstable by design
Local OS	Windows 11 (dev), Ubuntu (CI runners)
Node	Current LTS
Browsers — Playwright	Chromium, Firefox, WebKit
Browsers — Cypress	Chrome, Edge, Electron
CI	GitHub Actions: smoke on push/PR, full regression nightly (cron)
Reporting	Playwright HTML report + traces; Cypress Mochawesome HTML report; uploaded as CI artifacts

6. Roles and responsibilities

Single-contributor portfolio project; roles are worn as hats to mirror an enterprise setup:

ROLE	RESPONSIBILITY
QA Lead	Plans, designs (HLD/LLD), review gates, risk management, closure report
Automation Engineer	Page objects, specs, fixtures, CI pipelines, flake triage
Reviewer	PR review of test code against conventions (self-review checklist + async review)
Stakeholder (simulated)	Consumes smoke results per PR and the nightly regression trend

7. Risks and mitigation

#	RISK	LIKELIHOOD	IMPACT	MITIGATION
R1	Demo site resets data mid-run	High	High	Tests create their own data with unique suffixes; no reliance on prior state; retries on collision
R2	Shared tenant — other users mutate data concurrently	High	Medium	Unique naming per run; assert on records the test created, not on global counts
R3	Demo site slow or down	Medium	High	Generous but bounded timeouts; CI retry policy (1 retry); nightly failures auto-triaged before being treated as regressions
R4	OrangeHRM version upgrade changes DOM	Medium	High	Centralised locators in page objects; locator-preference rules; weekly triage
R5	Locator fragility (oxd-* generic classes, no test IDs)	High	Medium	Prefer role/label/text anchoring over class chains; document per-page locator decisions in LLD
R6	Dual-tool drift (Cypress vs Playwright coverage diverges)	Medium	Medium	Shared test-case IDs; parity check at each module's exit review
R7	Credentials leakage in repo	Low	High	.env + CI secrets; secret-scanning; demo creds are public but the habit must be enterprise-grade

8. Schedule (indicative, 6 weeks)

WEEK	MILESTONE
1	Requirement analysis + this test plan approved; environment setup validated (login green in CI)
2	Test case design complete for all modules; traceability matrix v1
3	Login + Dashboard automated (both tools), smoke pipeline live
4	Admin + PIM automated (both tools)
5	Leave automated; full regression nightly enabled; flake burn-down
6	Stabilisation, test summary report, retrospective, release tag

9. Deliverables

1. SDLC mapping (Artifact 1) and STLC mapping (Artifact 2)
 2. This Test Plan (Artifact 3)
 3. Framework HLD (Artifact 4) and module LLDs (Artifact 5)
 4. Manual test case catalogue with traceability matrix
 5. Cypress automation suite (JavaScript, POM)
 6. Playwright automation suite (JavaScript, POM)
 7. CI pipelines (PR smoke + nightly regression) with published HTML reports
 8. Defect log (GitHub Issues)
 9. Test Summary Report and retrospective notes
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10. Entry / exit criteria for the engagement

- **Entry:** demo app reachable, credentials valid, scope approved.
 - **Exit:** all five modules automated in both tools, flake rate < 2% across last 10 CI runs, summary report published, repo tagged and documented.
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